# VANI per-seller personas: built before the call, adapted live

PS04 Persona Design · IndiaMART Voice AI Hackathon · Sarvam agent **sulabh-im-80199a3e-82ea**

| 149,363 | 336,606 | 139,545 | 26 |
|---|---|---|---|
| sellers | past calls | transcript turns | live switches logged |

**Problem.** VANI talks to every seller in one voice and one style for the whole call. A rushed, irritated or confused seller still gets the same pitch, and meetings are lost.

**Solution.** A persona is generated **for each seller** from their own data and past calls: voice, language, tone and plan, each with a reason and evidence. VANI **adapts it live** on five signals and logs every switch.

## ARCHITECTURE AND DATA FLOW

### 1 · Data & evidence — offline, Python

**PS07 seller data** (CSV)
Profiles 149,363 · bot calls 269,455 · executive calls 67,151 · turns 139,545

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**Ingestion**: incremental, hash de-duplication, malformed and cp1252 rows handled. **Business names recovered** from call summaries (64,034 sellers); GLID is the key.

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**SQLite + SQLAlchemy** (WAL, indexed): suppliers, aliases, conversations, turns, call sessions, outcomes, switch log

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**Evidence miner**: meeting rate by pace, region, turnover, objection, callback history and VANI's turn after each signal. Every finding has *n* and a strength label.

→ seller record + evidence

### 2 · Persona API — FastAPI, Pydantic

**Persona engine** (rules, deterministic)
Voice · language · tone · plan (opening, personalisation, objection playbook, escalation, guardrails). Each has a **reason, data fields, evidence and confidence**; missing fields lower the confidence.

**Facts layer**: package suggestion by turnover (IndiaMART listed prices only); **numbers rendered in words** by code; greeting built per seller.

**Optional LLM**: gpt-4.1-mini or claude-haiku-5-5, structured output, injection guard, falls back to rules. **Off in this demo.**

**Endpoints**
`/supplier/persona` `/campaign/dial-check`
`/sarvam/pre-call` `/sarvam/switch-log`
`/sarvam/post-call` `/calls/{call_ref}`

**Safety**: bearer token, tunnel/proxy check, do-not-call stop variables, prompt-leak and fake-offer rejection

⇄ HTTPS tunnel · bearer token · 3 API tools

### 3 · Sarvam voice agent — live call

**BEFORE DIALLING**

`dial-check` removes do-not-call sellers (1,178 blocked)

**CALL START · ON_START**

**fetch_seller_persona** → 18 per-seller variables + unique `call_ref`. Greeting: *"Hello, kya aap {business} se bol rahe hain?"*

**DURING THE CALL · RUN**

**Sarvam LLM + female voice**, 11 languages, auto language detection, Hinglish. Prompt = `agent_prompt.md` + variables. Signals: frustration · confusion · rush · interest · language switch
→ change **delivery only**, never facts → **log_persona_switch**

**AFTER THE CALL · ON_END**

**send_call_outcome**: outcome, meeting/callback time, objections, summary → joined by `call_ref`

## SAMPLE PERSONA SPEC

**MS International** (6299) · New Delhi · trader, 5–25 Cr — Fatigued / sceptical

| Setting | Choice | Why: seller data → past-call evidence |
|---|---|---|
| Voice | Female, brisk pace (low) | Calls average 10 s · under-20 s sellers: 0.5% meetings vs 11.5% (n=45,242) |
| Language | Hindi → English, formal *aap* (high) | Delhi, Ltd company · Delhi calls ran in Hindi/Hinglish (n=40,301) |
| Tone | Warm, low pressure (low) | 4 "not interested" calls in 3 days, each under 15 s |
| Plan | One meeting ask → callback fallback · **close at the first sign of irritation** · IM Star Pro, listed price only | |
| Live rule | Every adapted turn ends with one clear question (after rush 16.8% vs 11.9%, n=1,957; after confusion 30.4% vs 23.9%, n=1,043) | |

## LIVE CALL WITH SWITCHES (SELLER 95487, TIME-PRESSED)

VANI: हेलो, क्या आप Kencraft & Allied Industries से बोल रहे हैं?

Seller: Busy hoon abhi, jaldi boliye. RUSH → one short question, two meeting times

Seller: Can we talk in English? What does it cost? LANGUAGE + INTEREST → English, "TrustSEAL Pro, sixty thousand rupees for one year"

Seller (6299): Roz roz call aata hai, time waste... FRUSTRATION → apologise, close politely

## BUSINESS IMPACT

- **More meetings:** each seller gets the opening, pace and objection handling that past calls show works for sellers like them.
- **Fewer burnt sellers:** fatigued sellers are closed at the first sign of irritation; do-not-call sellers are never dialled (1,178 blocked).
- **Trust and compliance:** the same listed prices, spoken in words, for every seller; no invented offers; truthful if asked about automation.
- **Measurable:** every switch and outcome is logged per `call_ref`, so the persona rules can be tuned on real results.
- **Proven:** all 5 required signals seen live · 5 clearly distinct sample personas · 106 automated tests · no LLM needed: rules explain every choice.

**Stack:** Python 3.10 · FastAPI · Pydantic v2 · SQLAlchemy 2 + SQLite · pytest · cloudflared · Sarvam Agents (MCP-configured) · optional OpenAI / Anthropic

**Personas:** Delhi (fatigued) · Guwahati (time-pressed, Bengali-ready) · Pune (established, Marathi-ready) · Amritsar (warm callback, Punjabi-ready) · Rajkot (value-seeking, Gujarati-ready)